

Future Paid Planning — Asset Maintenance Tool

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Status: Planning Document (Not Yet Implemented)

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Purpose: Document what requires paid services when revenue justifies it, with cost estimates and migration triggers

Executive Summary

This document outlines the **paid upgrade path** for each component of the Asset Maintenance Tool. The free-tier architecture (documented in `FREE_TIER_ARCHITECTURE.md`) is fully functional and production-ready. This document specifies **what to buy, when to buy it, and how much it costs** — so purchasing decisions are data-driven, not reactive.

Trigger for Paid Upgrades: Monthly Recurring Revenue (MRR) $\geq$ \$5,000 **OR** operational burden exceeds 20 hours/week.

Paid Upgrade Matrix

Component	Free (Current)	Paid Upgrade	Trigger	Est. Monthly Cost	Migration Effort
Database	Self-hosted PostgreSQL	Neon / Supabase / Timescale / RDS	MRR $\geq$ \$5k OR >50 tenants OR >100k assets	\$50-300/mo	Low (connection string swap)
Cache/Queue	Self-hosted Redis	Upstash / Redis Cloud / Valkey Cloud	MRR $\geq$ \$10k OR >500k jobs/mo	\$20-100/mo	Low (connection string swap)
Object Storage	MinIO	AWS S3 / Cloudflare R2 / GCS / Azure Blob	MRR $\geq$ \$5k OR >500GB storage	\$10-50/mo	Low (endpoint + creds swap)
Compute	Self-hosted VM	Fly.io / Railway / Render / Cloud Run / ECS Fargate	MRR $\geq$ \$10k OR >500 req/s	\$50-500/mo	Medium (Docker image deploy)
Monitoring	Prometheus/Grafana/Loki	Grafana Cloud / Datadog / Honeycomb	MRR $\geq$ \$20k OR >10 services	\$50-500/mo	Medium (agent install)

| ****SSL/CDN**** | Caddy + Let's Encrypt | Cloudflare Pro / AWS CloudFront | MRR ≥ \$5k ****OR**** >50k users | \$20-200/mo | Low (DNS + config) |

| ****Email**** | Self-hosted SMTP (Postfix) | SendGrid / Mailgun / SES / Postmark | MRR ≥ \$5k ****OR**** >10k emails/mo | \$15-100/mo | Low (SMTP creds swap) |

| ****SMS**** | N/A (not in free) | Twilio / Plivo / Vonage | MRR ≥ \$10k ****OR**** SMS feature launch | \$20-100/mo | Low (API key swap) |

| ****CI/CD**** | GitHub Actions (free tier) | GitHub Actions (larger runners) / CircleCI / Buildkite | MRR ≥ \$20k ****OR**** >2000 build min/mo | \$0-200/mo | Low (runner config) |

| ****Error Tracking**** | N/A (logs only) | Sentry / Bugsnag / Rollbar | MRR ≥ \$10k ****OR**** >100 errors/day | \$26-500/mo | Low (SDK install) |

| ****Search**** | PostgreSQL FTS | Meilisearch Cloud / Typesense Cloud / Algolia | MRR ≥ \$20k ****OR**** >1M records | \$50-500/mo | Medium (index sync) |

| ****Analytics**** | N/A (logs only) | Mixpanel / Amplitude / PostHog Cloud | MRR ≥ \$20k ****OR**** product analytics need | \$0-1000/mo | Medium (SDK install) |

Phased Upgrade Plan

Phase 1: First Revenue (\$1k-5k MRR)

****Priority:**** Reliability & Time Savings

| Upgrade | Cost | Why |

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| ****Managed PostgreSQL**** (Neon Pro / Supabase Pro) | \$50-100/mo | Eliminates DB ops burden; auto-backups, PITR, read replicas |

| ****Managed Redis**** (Upstash) | \$20-50/mo | Eliminates Redis ops; auto-scaling, multi-region |

| ****Cloudflare R2**** (or S3) | \$10-20/mo | Eliminates MinIO ops; global CDN, zero egress fees |

| ****Cloudflare Pro**** | \$20/mo | WAF, DDoS, Argo Smart Routing, cache analytics |

****Total Phase 1:**** ~\$100-190/mo

****Migration Time:**** 1-2 days (connection string swaps, DNS updates)

Phase 2: Growth (\$5k-20k MRR)

****Priority:**** Scale & Developer Velocity

| Upgrade | Cost | Why |

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| ****Managed Compute**** (Fly.io / Railway / Render) | \$100-300/mo | Auto-scaling, zero-downtime deploys, preview environments |

| ****Managed Monitoring**** (Grafana Cloud / Datadog) | \$100-300/mo | Pre-built dashboards, alerting, log correlation, 13-month retention |

| ****Error Tracking**** (Sentry) | \$26-100/mo | Real-time error tracking, release tracking, user impact |

| ****Email Service**** (SendGrid / Postmark) | \$20-50/mo | Deliverability, templates, analytics, suppression management |

| ****CI/CD Upgrade**** (GitHub Actions larger runners) | \$50-100/mo | Faster builds, macOS/ARM runners, more concurrent jobs |

****Total Phase 2:**** ~\$300-800/mo (cumulative: \$400-1000/mo)

Phase 3: Scale (\$20k-100k MRR)

****Priority:**** Enterprise Features & Compliance

| Upgrade | Cost | Why |

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| ****Managed Kubernetes**** (EKS/GKE/AKS) or ****ECS Fargate**** | \$500-2000/mo | Multi-region, auto-scaling, blue-green, compliance |

| ****Search Service**** (Meilisearch Cloud / Typesense) | \$100-500/mo | Sub-50ms search on 10M+ records, faceted search, typo tolerance |

| ****Product Analytics**** (PostHog Cloud / Mixpanel) | \$500-2000/mo | Funnel analysis, retention, feature flags, experimentation |

| ****CDN**** (CloudFront / Cloudflare Enterprise) | \$200-1000/mo | Global edge caching, WAF, DDoS, custom rules |

| ****Compliance**** (SOC 2 Type II audit, penetration test) | \$15k-50k one-time | Enterprise sales requirement |

****Total Phase 3:**** ~\$1000-5000/mo (cumulative: \$1500-6000/mo)

Phase 4: Enterprise (\$100k+ MRR)

****Priority:**** Multi-region, Compliance, Enterprise Sales

	Upgrade	Cost	Why
1	1. Upgrade to a more powerful processor (e.g., Intel Core i7 or i9)	~\$200 - \$400	Improves overall system performance, especially for multitasking and heavy applications.
2	2. Upgrade to a solid state drive (SSD)	~\$50 - \$150	Significantly reduces boot times and improves file access speeds.
3	3. Upgrade RAM to 16GB or 32GB	~\$80 - \$200	Allows for smoother multitasking and handling of memory-intensive tasks.
4	4. Upgrade the power supply unit (PSU) to a higher wattage, efficient model	~\$40 - \$100	Ensures stable power delivery to all components, especially under load.
5	5. Upgrade the cooling system (e.g., liquid cooling or high-quality fans)	~\$60 - \$150	Prevents overheating, which can lead to system instability and component degradation.
6	6. Upgrade the motherboard to a newer generation (e.g., Z790 for Intel)	~\$150 - \$300	Enables support for the latest processors, RAM, and other features.
7	7. Upgrade the graphics card (GPU) to a more powerful model	~\$200 - \$600	Improves gaming performance and rendering capabilities for professional applications.
8	8. Upgrade the case to a more spacious and well-ventilated model	~\$50 - \$100	Provides better airflow, making it easier to manage components and maintain lower temperatures.
9	9. Upgrade the operating system (e.g., Windows 11)	~\$100 - \$200	Provides the latest features, security updates, and improved compatibility with modern hardware.
10	10. Upgrade the network card to a faster model (e.g., 10GbE)	~\$50 - \$100	Enables faster network speeds for data transfer and online gaming.

A horizontal number line with four vertical tick marks. The segments between the tick marks are represented by dashed lines.

Multi-region Deployment \$2000-10000/mo Data residency (GDPR), disaster recovery, latency

Advanced Security (WAF, Bot management, API security) \$500-5000/mo Enterprise security requirements
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 Dedicated Support (Vendor enterprise plans) \$1000-10000/mo SLA, dedicated TAM, priority support

| ****Advanced Compliance**** (HIPAA, FedRAMP, ISO 27001) | \$50k-200k/yr | Regulated industry sales |

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Cost Projection Model

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Monthly Cost = Base Infrastructure + Per-Tenant Overhead + Usage-Based

Base Infrastructure (Phase 1): \$150/mo

Per-Tenant Overhead (est): \$2-5/tenant/mo

Usage-Based (storage, bandwidth): \$0.02/GB storage + \$0.09/GB egress

Example at 100 tenants, 50k assets, 500GB storage:

- Base: \$150
- Per-tenant: \$300 ($100 \times \3)
- Storage: \$10 ($500\text{GB} \times \0.02)
- Egress: \$45 ($500\text{GB} \times \0.09)
- Compute: \$200
- Monitoring: \$100
- Email/SMS: \$50

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Total: ~\$955/mo

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Migration Checklists

Database Migration (PostgreSQL → Managed)

- [] Provision managed instance (Neon/Supabase/Timescale)
- [] Configure VPC peering / private networking
- [] Run schema migration (same SQL, managed handles extensions)
- [] Set up read replicas for reporting
- [] Configure PITR (Point-in-Time Recovery) - 7-30 days
- [] Set up automated backups (daily + WAL archiving)
- [] Test PITR restore to staging
- [] Update connection strings in all services
- [] Configure PgBouncer on managed (or use managed pooler)
- [] Monitor for 48h before decommissioning self-hosted

Redis → Managed

- [] Provision managed Redis (Upstash/Redis Cloud)
- [] Configure TLS, ACLs, eviction policy
- [] Update BullMQ connection strings
- [] Test queue processing
- [] Configure Redis persistence (AOF/RDB)
- [] Set up Redis monitoring alerts

Object Storage (MinIO → S3/R2)

- [] Create bucket with same structure
- [] Configure CORS, lifecycle policies, versioning
- [] Set up bucket replication (if multi-region)
- [] Update application S3 client config (endpoint + creds)
- [] Migrate existing objects (mc mirror or rclone)
- [] Verify presigned URL generation
- [] Update CORS for direct browser uploads
- [] Test upload/download/delete flows
- ...

Compute (VM → Managed Platform)

- ...
- [] Containerize all services (Dockerfiles already exist)
- [] Push images to registry (GHCR / Docker Hub / ECR)
- [] Configure platform (Fly.io / Railway / Render / Cloud Run)
- [] Set up secrets management
- [] Configure autoscaling rules
- [] Set up health checks & readiness probes
- [] Configure custom domains + SSL
- [] Blue-green deploy to new platform
- [] Monitor for 48h before DNS cutover
- ...

Decision Framework

When to Upgrade Each Component

| Component | Hard Metric Trigger | Soft Signal Trigger |

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| ****Database**** | >100GB data OR >500 connections OR >50ms p99 query | "I spent 4h this week on DB maintenance" |

Redis	>1GB memory OR >10k ops/sec OR >50 queues	"Queue processing delayed during peak"
Storage	>500GB OR >10k requests/sec OR >\$50 egress/mo	"MinIO disk full alerts weekly"
Compute	>80% CPU for 1h OR >80% RAM OR >500 req/s	"Deploy takes 20min, can't preview PRs"
Monitoring	>10 services OR >1M metrics OR >100GB logs/mo	"Spent 2h debugging because no dashboard"
Email	>10k emails/mo OR deliverability issues	"Clients report missing notification emails"

Budget Planning Template

Month	MRR Target	Infrastructure Budget	Actual Spend	Components Upgraded
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1-3	\$0-1k	\$0	\$0	None (free tier)
4-6	\$1-5k	\$200		DB, Redis, Storage, CDN
7-12	\$5-20k	\$800		Compute, Monitoring, Email, CI/CD
13-24	\$20-100k	\$3000		K8s, Search, Analytics, Compliance
24+	\$100k+	\$10000		Multi-region, Compliance, Enterprise

Vendor Evaluation Criteria

Criterion	Weight
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Price at scale	30%
Migration ease	25%
Reliability/SLA	20%
Developer experience	15%
Data portability	10%

Evaluation Process:

1. Shortlist 3 vendors per component

2. Run 2-week proof-of-concept on staging
3. Score against criteria
4. Negotiate startup/enterprise discounts
5. Document decision in ADR

Cost Optimization Strategies (Even When Paid)

Strategy	Savings Potential
Reserved Instances / Savings Plans (1-3 yr)	30-60% compute/database
Spot/Preemptible Instances (batch jobs)	60-90% compute
Data Tiering (hot/warm/cold storage)	50-80% storage
Compression (Zstd for DB, gzip/zstd for HTTP)	20-50% bandwidth/storage
Query Optimization (indexes, materialized views)	50-90% DB CPU
Caching Strategy (stale-while-revalidate, CDN)	70-90% origin requests
Scheduled Scaling (scale down nights/weekends)	30-50% compute

Conclusion

The free-tier architecture is **production-ready and complete**. This document ensures that when revenue arrives, upgrading is:

1. **Predictable** — known costs, known triggers
2. **Incremental** — upgrade one component at a time
3. **Reversible** — same Docker images, portable config
4. **Data-driven** — metrics-based triggers, not guesswork

No premature optimization. Build on free tier until metrics demand paid upgrades. Then upgrade surgically, one component at a time.

End of Future Paid Planning Document